End-to-end utility versus cumulative secure time

CIFAR-10

CIFAR-100

Tiny ImageNet

TFE / ABY3

SecretFlow-SPU

8 releases / method

2 releases / method

18 releases / method

Development-validation accuracy

Cumulative online secure time (s, log scale)

Ours window Cumulative time (s)

Full Data window Cumulative time (s)

TFE Full Data

TFE Ours

SPU Full Data

SPU Ours

T@80/90/95

I-bars show sample standard deviation for TFE n=3; Tiny TFE and all SPU panels are n=1. SPU uses broken absolute cumulative-time windows so every genuine 8/2/18 release is visible; backend rows are not hardware-normalized.